Inayat **Ali**, PhD

Postdoctoral Researcher | Computer Systems, Networked Systems & Edge-Cloud Computing
Daejeon, Republic of Korea | Ph.D. in Data & HPC Science | Senior Member, IEEE
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Academic Profile

Computer systems and networking researcher with **10+ years of research experience** across government-funded research institutes in **South Korea** and higher education. Currently a Postdoctoral Researcher at ETRI, South Korea, with an active research line spanning low-latency transport, congestion control, programmable networks, and Kubernetes-based edge-cloud resource orchestration. Experienced in higher-education teaching, graduate research mentoring, international collaboration, and applied systems R&D.

Academic Highlights

- **Research:** 10+ years across ETRI, KISTI, and higher education, with an active line in computer systems, AI-based intelligent networking, and edge-cloud orchestration.
- **Teaching & mentoring:** Taught four undergraduate courses and mentored three Ph.D./Master's researchers; mentoring led to three peer-reviewed journal papers as corresponding author and several under review.
- **Leadership & funding:** Co-developed a successful three-year Korean government-funded proposal; international collaborations span Korea, Pakistan, Cyprus, and Turkey, with six journal manuscripts currently under review.
- **Innovation & impact:** Two granted/registered patents, one published patent application, and contribution to one ETRI technology-transfer activity.

Research Interests

Computer Systems	Distributed and networked systems; cloud/edge computing; AI data-center networks and RDMA-based high-performance networking; resource orchestration and auto-scaling; high-performance and low-latency systems.
Networks & Protocols	Transport protocols; congestion control and AQM; programmable networks; SDN/P4; future Internet; NDN/ICN
AI for Systems	Learning-assisted network optimization; performance-aware resource allocation; intelligent orchestration for latency-critical applications; AI for Network security

Academic and Research Appointments

Electronics and Telecommunications Research Institute (ETRI)
Daejeon, South Korea
Postdoctoral Researcher — *May 2022 – Present*

- ETRI is **Korea's largest government-funded research institute specializing in information and communication technology (ICT)**, with a long record of national-scale technology R&D and technology transfer.
- Research and develop low-latency Internet transport, congestion control and AQM, programmable networking, and cloud/edge resource orchestration for latency-critical applications.
- Develop learning-assisted network-control mechanisms and Kubernetes-based auto-scaling approaches, and conduct hands-on experimentation using clusters, Linux servers, network switches, simulation/emulation platforms, and performance-monitoring tools.
- Contribute to Korean national R&D projects and research outputs including peer-reviewed publications, patents, prototypes, and a team technology-transfer activity.

Korea Institute of Science and Technology Information (KISTI)
Daejeon, South Korea
Graduate Student Researcher — *Sept. 2017 – Feb. 2022*

- Worked full-time at KISTI while pursuing the UST Ph.D., conducting hands-on research in Named Data Networking, vehicular networking, mobility management, and scientific-data delivery.
- Designed and evaluated protocol mechanisms for content retrieval, mobility support, and broadcast mitigation, leading to peer-reviewed publications and granted patents.
- Co-developed a successful research proposal that received three years of Korean government research funding.

Comwave Institute of Sciences and Information Technology
Abbottabad, Pakistan
Visiting Lecturer — *Sept. 2015 – Jan. 2017*

- Taught undergraduate computer science students in *Object-Oriented Programming*, *Computer Networks and Communications*, and *Introduction to Telecommunication Systems*.
- Developed curriculum for an undergraduate course that aligned with industrial needs.

COMSATS University of Science and Technology
Abbottabad, Pakistan
Research Associate — *Sept. 2015 – Jan. 2017*

- Conducted research in programmable networks and Software-Defined Networking while contributing to undergraduate computer science teaching.
- Combined academic teaching with research on policy-based routing and network programmability, establishing an early research-teaching connection in computer systems and networking.

Education

Korea University of Science and Technology (UST), KISTI School — *Daejeon, South Korea*

Ph.D. in Data & HPC Science (Computer Networks) | Master-PhD Combined Program — *Sept. 2017 – Feb. 2022*

- **CGPA:** 4.15/4.5
- **Dissertation:** "Efficient Content Retrieval in Named Data Networking Based Highly Dynamic Wireless Environment."
- UST is *Korea's only government-funded research-institute university*; its education model integrates on-site research at government-funded research institutes, participation in national R&D projects, and access to advanced research facilities. Ph.D. research was conducted full-time at the KISTI School.

COMSATS University of Science and Technology — *Abbottabad, Pakistan*

B.S. in Telecommunication and Networking | Four-Year Honors Degree — *Feb. 2011 – Mar. 2015*

- **CGPA:** 3.61/4.0
- **Distinction:** Academic Silver Medal
- **Undergraduate Thesis:** "Policy Based Routing Using Software Defined Networking (SDN)."

Teaching, Graduate Mentoring and Problem-Based Learning Contribution

- **Higher-education teaching:** Taught *Object-Oriented Programming*, *Computer Networks and Communications*, *Advanced topics in computer networks*, and *Introduction to Telecommunication Systems* to undergraduate students, providing experience across core programming, networking, and communication-system topics relevant to Bachelor-level computer science.
- **Problem-Based Learning (PBL) alignment:** A decade of hands-on research in government-funded research institutes has centered on open-ended, real-world systems problems requiring collaborative problem formulation, independent investigation, prototyping, experimentation, and evidence-based discussion. This experience is well aligned with Maastricht University's constructive, contextual, collaborative, and self-directed PBL principles and can be translated into small-group tutorials, practicals, skills training, and project-centred computer-systems learning.
- **Graduate research mentoring and student guidance:** Mentored **three Ph.D.- and Master's-level researchers** in South Korea and Pakistan, guiding problem formulation, methodology, experimentation, analysis, manuscript development, and publication; this work led to **three peer-reviewed journal publications** for which I served as corresponding author and several are under review.
- **Curriculum contribution:** Current applied R&D experience in Kubernetes, cloud/edge systems, programmable networks, AI-assisted systems optimization and security, low-latency transport, Linux servers, physical switches, P4, NS-3, and Mininet can support core teaching, hands-on projects, and systems-oriented electives while connecting foundational concepts to contemporary computing practice.

Research Leadership, Funding and International Collaboration

- **Competitive funding experience:** Co-developed the research concept, technical methodology, and work plan for a proposal with my Ph.D. supervisor; the successful proposal received **three years of Korean government research funding**.
- **International collaboration:** Built research collaborations with researchers in **South Korea, Pakistan, Cyprus, and Turkey**; these collaborations have produced published work and **five journal manuscripts currently under peer review**.
- **Applied research:** Contribute to multidisciplinary national R&D at ETRI spanning computer systems, intelligent networking, resource orchestration, intellectual-property generation, and technology transfer.

Technical and Experimental Expertise

Systems	Kubernetes clusters, Linux servers, network switches, containerized systems, cloud/edge orchestration
Programming	Python, C++, C#, network programming, P4
Experimentation	NS-3, Mininet, ndnSIM, Packet Tracer, systems/network performance measurement
Networking	TCP/IP, congestion control, ECN, AQM, SDN, NDN/ICN, transport protocols, programmable networks
Languages	English (Proficient; IELTS C1, band 7), Urdu (Native), Korean (Beginner)

Patents and Technology Transfer

- **U.S. Patent - Granted.** H. Lim and **I. Ali**, "Forwarding method and system for data broadcast reduction based on name centricity and applied in vehicle named data networking." **U.S. Patent No. 12,464,588 B2**; Publication No. US 2023/0217522 A1; publication date: Jul. 6, 2023; granted: Nov. 2025.
  Patent link: `https://patents.google.com/patent/US12464588B2/en`
- **Korean Patent - Registered.** H. Lim and **I. Ali**, "Name-Centricity-Based Broadcast Mitigation Forwarding Method and System Applied in VNDN." **Application No. 10-2021-0192759**; **Registration No. 10-2668825**; registration date: May 20, 2024.
  Patent link: `https://doi.org/10.8080/1020210192759`
- **Korean Patent Application - Published.** P. Park, B. Kwak, M. Seong, **I. Ali**, Y. H. Sun, and S. W. Hong, "Apparatus and Method for Congestion Control Mechanism Based on Reinforcement Learning to Support Ultra-Low-Latency Interactive Services in Communication System." **Application No. 10-2024-0001562**; application date: Jan. 4, 2024; **Publication No. 10-2025-0106938**; publication date: Jul. 11, 2025.
  Patent link: `https://doi.org/10.8080/1020240001562`
- **Technology transfer:** Contributed to the technology transfer of an ultra-low-latency transport protocol for time-sensitive 5G/6G applications, incorporating application-aware QoS APIs, deadline-based retransmission and scheduling, and object-level processing for V2X, VR/AR/XR, interactive holograms, and other real-time services.

Selected Research Projects

- **Resource Orchestration and Auto-scaling for Guaranteed Application Performance** — ETRI, South Korea (Apr. 2025–Present). Develop intelligent, policy-driven Kubernetes orchestration and auto-scaling using workload, queueing, latency, and resource-performance information to maintain application-level performance guarantees.
- **Design of High-Performance, Low-Latency Internet Transport Protocol** — ETRI, South Korea (May 2022–Present). Design, prototype, simulate, and evaluate transport and congestion-feedback mechanisms for latency-critical and interactive applications.
- **Efficient Content Retrieval and Mobility Management in Vehicular Named Data Networking** — KISTI, South Korea (Sept. 2017–Feb. 2022). Designed and evaluated mobility-management and broadcast-mitigation mechanisms for NDN-based vehicular environments, resulting in publications and patents.

Professional Service

- **Journal Reviewer:** IEEE Internet of Things Journal; IEEE Access; Computer Communications; IEEE Transactions on Network Science and Engineering; Journal of Supercomputing; Scientific Reports; Computers & Electrical Engineering.
- **Review record:** ORCID `https://orcid.org/0000-0001-8847-294X`

Honors and Awards

- **2026 - IEEE Senior Member:** I was recently elevated to the IEEE Senior Member position.
- **2017 - Ph.D. Scholarship**, Korea University of Science and Technology (UST), South Korea.
- **2016 - Graduate Scholarship**, COMSATS University of Science and Technology, Pakistan.
- **2015 - Academic Silver Medal**, B.S. Graduation, COMSATS University of Science and Technology, Pakistan.
- **2014 - Second Position**, CS Project Competition, COMSATS Islamabad, Pakistan.